

Modified Entringer Connection Constants and Spectral Limits

Exact amplitude, all fixed orders, inverse bounds,
and a marked Bernoulli limit for OEIS A386381 and A386363

Research report 180

3 October 2026

Abstract

The diagonal d_n of the modified Entringer triangle A386363 is A386381. Its leading asymptotic scale and numerical amplitude were already recorded in the OEIS. We identify that amplitude exactly as $c = \pi A(\pi/2)$, where $(zA')' = (\sec z + \tan z)A$ and $A(0) = A'(0) = 1$, and give an outward certified decimal enclosure. A resonant Frobenius expansion supplies an explicit recurrence for every fixed asymptotic order. A Lambert W approximation and finite-order carriers give controlled eventual inverse brackets, without an unconditional rounding rule.

Marking the recursive boundary injections gives an entire endpoint function $C(\lambda)$ and an absolute, compact-complex-uniform expansion. We prove that $C(\lambda)$ is the Fredholm determinant of a positive trace-class Green operator on a space with infinite underlying measure. The limiting coefficient law is a countably infinite sum of independent Bernoulli variables; convergence has all fixed-order exponentially weighted ℓ^1 corrections. Large positive marks also yield an explicit far-tail law for the limiting count. The limiting zeros are simple and negative. Every finite nonreal zero escapes bounded sets, although an exact degree-eight example disproves finite real-rootedness.

The boustrophedon transform, singularity analysis, Fredholm theory and Bernoulli spectral interpretation are classical. The report gives family-specific deductions with full proofs and bounded source comparison, not a historical priority claim. The asymptotic remainder constants are non-effective; the amplitude certificate is a separate finite calculation with explicit tail bounds.

[write] **Status and trust boundary, added 7 October 2026, batch 110.** This is bundle Report 180 of an external AI-assisted research session, filed in ProveIt as the single-source research report **a386381-modified-entringer** (provenance, the sources as the write read them, what was checked, relation to the repository, the collected non-claims and the reading conventions are in Section 1.1). It is unrefereed and not formalized. Its author line reads “Research report 180”: it names no person or tool. *What rests on what.* Proposition 2.1 (the boustrophedon reduction, credited to Millar, Sloane and Young and also derived directly), the Frobenius construction, Theorems 5.1, 8.1, 9.1, 9.2, 10.1, 10.3 and 11.1 and the inverse statements have conventional proofs that use no computation. The decimal enclosure of the amplitude (36) combines proved Taylor tails with interval arithmetic in mpmath, whose implementation is trusted; the coefficient tables, the Sturm count for (74) and the displayed correction polynomials are exact finite computations of the shipped programs. Every other decimal is a diagnostic. The recurrence, the leading scale and the numerical amplitude 25.5745... are prior (OEIS; Kurkov, Kotěšovec); the new content is the exact identification of the amplitude as a connection value and everything built on it. The write adds Remark 1.1 (the OEIS entries quoted, and all 265 b-file terms recomputed), Remark 7.2 (the inverses against the transseries volume, statement by statement), and dated notes. No statement, proof or number of the source was changed.

Contents

1	The sequence and the precise scope	3
2	From the triangle to a differential equation	6
3	Boundary marking and an entire endpoint function	7
4	Frobenius construction uniform in complex marks	9
5	All fixed asymptotic orders with absolute uniform errors	10
6	An exact Wronskian formula and a certified amplitude	12
7	Logarithmic carriers and discrete inversion	14
8	A Green operator despite the infinite measure	16
9	The limiting law and every fixed probability correction	17
10	Growth for a large positive boundary mark	19
11	Simple limiting zeros and finite nonreal zeros	24
12	Computational evidence and reproducibility	25
13	Source limits and further directions	26

1 The sequence and the precise scope

Let $T(n, k)$ be the triangle specified in OEIS A386363 [1]:

$$\begin{aligned} T(0, 0) &= 1, & T(n, 0) &= 0 \quad (n > 0), \\ T(n, 1) &= T(n-1, n-1) \quad (n > 0), \\ T(n, k) &= T(n, k-1) + (n-2)T(n-1, n-k) \quad (2 \leq k \leq n). \end{aligned} \tag{1}$$

Write $d_n = T(n, n)$, the sequence A386381 [2]. The first terms are

$$\begin{aligned} &1, 1, 1, 2, 8, 56, 640, 10960, 264640, 8581760, 360331520, 19031302400, \\ &1235451750400, 96722377139200, 8988790940876800, 978442125179648000, \\ &123324448870740377600, 17820979140159760793600, \\ &2926936219425738642227200, 542215853077506417192140800, \\ &112527512540808439576566169600. \end{aligned}$$

The definitions and recurrence are prior work. The triangle entry is attributed to Mikhail Kurkov, 19 July 2025. The diagonal entry attributes the numerical leading equivalent

$$d_n \sim c \frac{2^{n+1} n^{2n-3}}{e^{2n\pi n-1}}, \quad c = 25.574519628957467521537232312735336894\dots, \tag{2}$$

to Václav Kotěšovec, 2 September 2025. We neither present that numerical observation as new nor infer from the absence of a linked proof that no proof exists elsewhere.

Throughout put

$$\rho = \frac{\pi}{2}, \quad E(z) = \sec z + \tan z = \tan \left(\frac{\pi}{4} + \frac{z}{2} \right), \quad N = n - 2.$$

The distinction between N and n matters: the natural factorial carrier uses N , while the elementary logarithmic carrier and inverse use n .

An “all fixed orders” expansion means that the truncation order is fixed before $N \rightarrow \infty$. Its constants may depend on that order and specified compact parameter sets. No convergence of the complete inverse-power series, growing-order error theorem, complete collection of exponentially small sectors, or explicit asymptotic onset is asserted. The marked statistic is defined by the recursive polynomial array below. No external combinatorial bijection is assumed.

The classical transform theorem of Millar, Sloane and Young [3] supplies the central exact reduction; we also derive it in the present normalization. Frobenius analysis, Abel’s identity, logarithmic singularity transfer [4, 5], Lambert inversion and trace-class determinant methods are established tools. The Bernoulli interpretation is the familiar spectral phenomenon described by Hough, Krishnapur, Peres and Virág [6]. Our direct product proof requires no construction of a point process.

Remark 1.1 ([write] The OEIS entries, quoted, 7 October 2026). The entries were read on 7 October 2026 in the OEIS internal format; the quotations are verbatim.

- (a) A386363 [1] (revision #24, 27 February 2026), offset 0, author line “_Mikhail Kurkov_, Jul 19 2025”, is named

Variation of triangle of Entringer numbers (A008281) read by rows: $T(n, k) = T(n, k-1) + (n-2)*T(n-1, n-k)$ for $1 < k \leq n$, $T(n, 1) = T(n-1, n-1)$ for $n > 0$, $T(n, 0) = 0^n$.

which is (1). Its example rows agree with the write’s own computation of the triangle (for instance row 8 is 0, 10960, 76720, 140800, 196480, 237760, 260800, 264640, 264640).

- (b) A386381 [2] (revision #27, 27 February 2026), offset 0, “Main diagonal of A386363.”, author line “_Mikhail Kurkov_, Jul 20 2025”. Its asymptotic line reads

$$a(n) \sim c * 2^{(n+1)} * n^{(2*n-3)} / (\exp(2*n) * \text{Pi}^{(n-1)}), \text{ where } c = 25.574519628957467521537232312735336894\dots$$

- _Vaclav Kotesovec_, Sep 02 2025

which is (2). Corollary 5.2 proves it, with $c = \pi C(1)$; the printed decimal is the truncation (not a rounding) of the enclosure (36). The entry links a b-file “Table of n , $a(n)$ for $n = 0..264$ ” (Kotěšovec). Its other formula lines, Kurkov’s Conjectures 1 and 2 on divisibility (the second concerns the Euler numbers A000111) and Luschny’s conjecture that $2a(n)$ is divisible by A060818(n), are outside the scope of this report and are not addressed by it.

The b-file, fetched on 7 October 2026, has 265 terms; the write recomputed $d_0, \dots, d_{264}$ from (1) in exact integer arithmetic with its own code: all 265 agree. (Section 13 records that the source could not retrieve this file.) OEIS content is published under CC BY-SA 4.0. Nothing was submitted to the OEIS.

1.1 Provenance, sources and reading conventions

[write] **Added 7 October 2026, batch 110.** *Provenance.* This report is Report 180 of the session bundle that ProveIt’s docs/incoming intake processed as batch 110: the archive `Modified_Entringer_Asymptotics_and_Spectral_Limits_Source.zip` (789,613 bytes, SHA-256 23d6b43bbad1...e089a7f571f95f, 34 files, no wrapper directory), arrival commit 60f54ea06, placed in 8622ca7e5. The text is the delivered `Report180.tex` (1,018 lines, dated 3 October 2026; the delivered PDF has 22 pages), shipped as `article.tex`. The exact core is shipped as `code/verify.py`, `code/exact.py` and `code/regenerate.py`; the delivered root programs `build.py`, `guard_tests.py`, `test_build.py` and `verify_manifest.py` under `code/`; the optional programs as `code/optional-*.py`; the frozen references as `data/references-*.json`, the generated summaries as `data/generated-*.json`, and `README_CODE.md`, `SOURCE_AUDIT.md` and `optional-README.md` at the report root (the report README maps every name). Not shipped, and retrievable from the arrival commit: the delivered PDF, the checksum manifest `SHA256SUMS.json` (33 entries, verified at placement) and the delivered README, replaced by the report README. The package records no ProveIt commit; its source audit names three neighbouring reports by their GitHub paths. It carries no “prepared for private review” line and no personal data. It is a single-source report, so no merge choice was made. The write prefixed the 102 delivered labels with `men:` and updated the 73 references to them; it added labels to Section 1 and to this subsection, the notes marked [write], Remarks 1.1 and 7.2, and dated notes. The added remarks are the last statements of their sections, this subsection follows the last delivered text of Section 1, and the added displays are unnumbered, so every theorem, section and equation of the source keeps its delivered number.

The sources, as the write read them (7 October 2026).

- OEIS A386363 and A386381 with the b-file of A386381, in the internal format (Remark 1.1).
- The transseries volume `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`: `p0:def:core`, `p0:prop:factorial-core`, `p0:def:three-inverses` and `p0:thm:staircase`, read for Remark 7.2.
- Not read by the write: the boustrophedon paper [3] (the identity used, $B = EA$ for exponential generating functions, is also derived directly in the proof of Proposition 2.1), the singularity-analysis sources [4, 5], HKPV [6] and the DLMF sections [7]; the source’s account of them is reported, not confirmed.

What was checked at intake. The batch-110 dossier read the manuscript and found no error. On copies it ran `code/verify.py` in normal and optimized mode (identical output, equal to the recorded

generated/verification.json: 21 OEIS terms, exact $n \leq 101$, corrections through order six, six real and two nonreal roots of P_{10}) and **code/regenerate.py** with `-compare` (pass). *What the write checked* (Python 3.14.4, SymPy 1.14.0, mpmath 1.3.0, own code). Every proof was read line by line, with no error found; in particular the monotonicity step $d_n \geq (n-1)d_{n-1}$ in the proof of Theorem 7.1 (it uses $T(n-1, n-2) = T(n-1, n-1)$ for $n \geq 3$), the Stirling step of Corollary 5.2, and the equivalence of (63) and (64) through $(2m+1)! = 4\sqrt{\pi} m^{3/2} (2m/e)^{2m} \{1 + O(m^{-1})\}$. By exact symbolic re-expansion from the recurrences (15) and (18) it recomputed $L_0(1), \dots, L_4(1)$ of (23), the coefficient of t^4 in H_1 , the six corrections (29), the polynomials $s_1, \dots, s_4$ of (27), Q_1, Q_2, Q_3 of (49), the expansion (52), the mean and variance corrections (53)–(54), and the coefficients $1/6, -2, \rho^2/3 - 541/180$ of (38): all agree. It recomputed P_{10} from the marked array (8) (equal to (74), with $P_{10}(1) = d_{10}$), its six real roots and its trivial gcd with its derivative; $C(1)$ from the midpoint formula (31) at $m = 420$ and 70 digits (note at the end of Section 6); the optical length $\mathcal{A} = 4.2586174557058935695\dots$; and $C(\lambda)$ for large λ (note after the proof of Theorem 10.1). On a copy of the delivered layout rebuilt from the shipped files, the verifier and the regeneration comparison pass with output byte-identical to the dossier’s run. Floating computations are observations, not certificates.

Relation to the repository. Before batch 110 no file of the repository named A386381 or A386363. No statement of this report is formalized in Lean or Rocq, and its place in the collection confers no formal status. Nearest reports, all in `SetTheory/Cardinals/docs/reports/`: `a125054-central-poupard-numbers` (tangent numbers by the Seidel boustrophedon, pole-sector asymptotics), `congruences-and-valuations/secant-number-periodicity` (secant numbers through the Entringer triangle, congruences), `a205497-zigzag-eulerian-spectra` and `a122399-surjection-diagonal`. They share classical ingredients but no object or result with this report, so no reciprocal note is proposed. The inverses are compared with the transseries volume in Remark 7.2.

[write] What is not claimed, collected from the source. The recurrence, the leading scale and the numerical amplitude are prior (OEIS), and the source does not “present that numerical observation as new”. It makes “not a historical priority claim”: the boustrophedon transform, singularity analysis, Fredholm theory and the Bernoulli spectral interpretation are classical, and its overlap review is bounded (“Search misses ... are not evidence of worldwide novelty or exhaustive non-overlap”); it settles no posed open problem. Every expansion is for a fixed order; no convergence of the inverse-power series, growing-order error theorem, complete set of exponentially small sectors or explicit asymptotic onset is asserted; the remainder constants are non-effective. The marked statistic is the recursive coefficient statistic, with no claimed combinatorial family. The amplitude certificate trusts mpmath’s interval implementation and is “not a proof-assistant verification”; the longer decimal strings are diagnostics, and the enclosure gives no effective finite- N constants. The inverse bracket is not an effective finite-input certificate, and $\lceil x_1 \rceil$ or $\lceil \xi_M \rceil$ is not asserted to work. Theorem 10.1 gives no complex-sector uniformity and no joint limit with N ; the tail results concern the limiting law only, with non-effective constants and no conditioned-Poisson law. Theorem 11.1 is a bounded-region statement, with no eventual real-rootedness and no escape rate. The write adds: its checks are exact finite or floating computations; it claims no novelty for any inversion (Remark 7.2).

Reading conventions (letters the source reuses, with the tempting false readings; no symbol of the source was renamed; symbols of the transseries volume carry the subscript “vol” in Remark 7.2).

Symbol	Meaning here (where)	Not to be read as
$c, C(\lambda), C_*$	the amplitude $\pi C(1)$ (14); the endpoint $A_\lambda(\rho)$; $\log(2\pi c)$ (37); $C_{R,M}, c_0, c_k$ are generic constants	each other
$A, A_\lambda, \mathcal{A}$	the scalar solution A_1 ; the marked solution; the optical length (55)	each other; the volume’s A_n , $\mathcal{A}_{\text{vol}}$ (an interpolation)
d, d_n	$1/(e\sqrt{\rho})$ (37); the sequence A386381	each other; the volume’s d_{vol}
$E, \mathbb{E}, e_n$	$\sec z + \tan z$; expectation; Taylor coefficients of E	each other; Euler’s e
$K, K_0, K_M, K_{\text{tail}}, k$	the Green operator (Section 8) but a compact mark set in Theorems 5.1 and 9.2; the trace (12); the inverse constant; the tail constant (62); the kernel $k(s, t)$, but $k = \sqrt{\lambda}$ in Section 10	each other

Symbol	Meaning here (where)	Not to be read as
$L, L_j, L_\lambda, \mathcal{L}_\lambda$	$\log X$; the coefficients (21); the series $\rho \cot(\rho t/2) H_\lambda$; the differential operator (16)	each other; the volume's L_{vol}
$H_\lambda, H(x)$	the Frobenius solution (Lemma 4.1); the tail carrier of Corollary 10.4	each other
$G_\lambda, G, G_N, \mathcal{G}$	the second Frobenius germ; the limiting PGF (46); the finite PGF (9); the space $L^2(dx/x)$	each other
$W(r, j), W_0, W_x$	the normalized array (3); Lambert's function; a Wronskian	each other
$p_n, p_j, p(m), q_n, q_m$	Taylor coefficients of A (7); Bernoulli parameters (46); masses $\mathbb{P}(J = m)$; $p_n \rho^n$; $\mathbb{P}(J \geq m)$	each other
Q_λ, Q_j, Q	the analytic part in (22); the PGF corrections (48); the Liouville potential	each other
N, n, M, m	$N = n - 2$ (the factorial carrier) and n (the logarithmic carrier and inverse); a fixed order, but $M(r)$ the tilted mean; the truncation degree $m = 420$, but m a mass index in Section 10.1	each other
$V, V(r), v_J$	the Volterra operator (11); the tilted variance; the limiting variance	each other
S, μ, μ_J	the operator (43), but a Bernoulli sum in (69); the measure $E(s) ds$; the limiting mean	each other
x_0, x_1	Lambert starts for d_n (39); for the tail inverse (71)	each other
$B, B_{2j}, B_\nu, B_1, B_2$	the boundary series; Bernoulli numbers; $\sup y I_\nu K_\nu$; generic constants	each other
ρ, κ_j, Φ	$\pi/2$; eigenvalues of K ; $\rho t \cot(\rho t/2)$	the volume's $\rho_{\text{vol}}, \kappa_{\text{vol}}, \Phi_0$
$T(n, k), T, t$	the triangle; $\log(1/y)$ in Corollary 10.4; $1 - z/\rho$, also an integration variable	each other
δ_m, δ	the midpoint error bound (35); $\rho - z$, and the secant step, in Section 10	each other

2 From the triangle to a differential equation

For $n \geq 2$, the last two entries in row n coincide because the final increment in (1) is $(n - 2)T(n - 1, 0) = 0$. Define

$$W(r, j) = \frac{T(r + 2, j + 1)}{r!} \quad (0 \leq j \leq r), \quad a_r = W(r, 0), \quad b_r = W(r, r). \quad (3)$$

Then $a_0 = b_0 = 1$, and direct substitution gives

$$W(r, j) - W(r, j - 1) = W(r - 1, r - j), \quad a_r = \frac{b_{r-1}}{r} \quad (r \geq 1). \quad (4)$$

The factorial normalization need not be integral. Set

$$A(z) = \sum_{r \geq 0} a_r \frac{z^r}{r!}, \quad B(z) = \sum_{r \geq 0} b_r \frac{z^r}{r!}.$$

Proposition 2.1. *The selected analytic solution at zero satisfies*

$$B = EA, \quad (zA')' = EA, \quad A(0) = A'(0) = 1. \quad (5)$$

Moreover,

$$d_{N+2} = (N!)^2 [z^N] B(z) \quad (N \geq 0). \quad (6)$$

Proof. The identity $B = EA$ is the boustrophedon transform theorem [3, Theorem 1]. To verify it directly, introduce the bivariate exponential series

$$F(x, y) = \sum_{r \geq 0} \sum_{j=0}^r W(r, j) \frac{x^j y^{r-j}}{j!(r-j)!}.$$

The difference recurrence is exactly $F_x - F_y = F(y, x)$. With $u = x + y$ and $v = x - y$, write $\tilde{F}(u, v) = F((u+v)/2, (u-v)/2)$. Then $2\tilde{F}_v(u, v) = \tilde{F}(u, -v)$ and hence $4\tilde{F}_{vv} = -\tilde{F}$. At $v = 0$ the first equation equates the sine and cosine coefficients. Therefore

$$\tilde{F}(u, v) = D(u) \{ \cos(v/2) + \sin(v/2) \}.$$

The original boundary values are $F(0, z) = A(z)$ and $F(z, 0) = B(z)$, so

$$\frac{B(z)}{A(z)} = \frac{\cos(z/2) + \sin(z/2)}{\cos(z/2) - \sin(z/2)} = E(z).$$

This is a formal identity near zero; the denominator has constant term one. The relation $a_{r+1} = b_r/(r+1)$ gives $(zA')' = B$. Finally $d_{N+2} = N!b_N$ by (3), yielding (6). $\square$

For an exact scalar coefficient algorithm write $E(z) = \sum e_n z^n$ and $A(z) = \sum p_n z^n$. Since $E' = (1 + E^2)/2$,

$$\begin{aligned} e_0 &= p_0 = 1, \\ 2(n+1)e_{n+1} &= \sum_{j=0}^n e_j e_{n-j} + \mathbf{1}_{\{n=0\}}, \\ (n+1)^2 p_{n+1} &= \sum_{j=0}^n e_j p_{n-j}. \end{aligned} \quad (7)$$

All coefficients are positive rationals. These recurrences and the original integer triangle provide independent finite computations of the same diagonal.

3 Boundary marking and an entire endpoint function

The marking starts after the fixed $n = 2$ base. Define integer polynomials $U_{r,j}(\lambda)$ by

$$\begin{aligned} U_{0,0}(\lambda) &= 1, \\ U_{r,0}(\lambda) &= \lambda U_{r-1,r-1}(\lambda), \\ U_{r,j}(\lambda) &= U_{r,j-1}(\lambda) + r U_{r-1,r-j}(\lambda) \quad (1 \leq j \leq r), \end{aligned} \quad (8)$$

and put $P_{r+2}(\lambda) = U_{r,r}(\lambda)$. At $\lambda = 1$ this is precisely $T(r+2, j+1)$ and $P_n(1) = d_n$. Coefficients are nonnegative integers. The exponent of λ records how many left-boundary injections occur in the recursive expansion, counted with the recurrence multiplicities. Equivalently, the following probability definition needs only these coefficients:

$$\mathbb{P}(J_N = k) = \frac{[\lambda^k] P_{N+2}(\lambda)}{P_{N+2}(1)}, \quad G_N(\lambda) = \mathbb{E} \lambda^{J_N} = \frac{P_{N+2}(\lambda)}{P_{N+2}(1)}. \quad (9)$$

This is a valid finite law, with no claim that J_N is a statistic in an independently identified classical combinatorial family.

Apply the transform proof to $W_\lambda(r, j) = U_{r,j}(\lambda)/r!$. Its boundary relation has the extra factor λ , so

$$B_\lambda = EA_\lambda, \quad (zA'_\lambda)' = \lambda EA_\lambda, \quad A_\lambda(0) = 1, \quad A'_\lambda(0) = \lambda, \quad (10)$$

and $P_{N+2}(\lambda) = (N!)^2[z^N]B_\lambda$. Integrating twice on the real interval gives

$$A_\lambda(z) = 1 + \lambda(VA_\lambda)(z), \quad (Vf)(z) = \int_0^z E(s)f(s) \log \frac{z}{s} ds. \quad (11)$$

The integral at $s = 0$ is interpreted as an improper integral.

Theorem 3.1. *The endpoint*

$$C(\lambda) = A_\lambda(\rho) = \sum_{j \geq 0} \lambda^j (V^j 1)(\rho)$$

is an entire function of λ . If

$$K_0 = \int_0^\rho E(s) \log \frac{\rho}{s} ds, \quad (12)$$

then

$$0 \leq (V^j 1)(\rho) \leq \frac{K_0^j}{j!}, \quad |C(\lambda)| \leq e^{|\lambda|K_0}, \quad 1 < C(1) \leq e^{\pi^2/3} < 200. \quad (13)$$

Proof. Near zero the integrand in (12) is $O(\log(1/s))$. Near ρ , the simple pole $E(s) \sim 2/(\rho - s)$ is cancelled by $\log(\rho/s) \sim (\rho - s)/\rho$. Thus $K_0 < \infty$.

In the ordered integral for $V^j 1$, every factor $\log(s_{i+1}/s_i)$ is bounded by $\log(\rho/s_i)$, with the endpoint interpreted as $s_{j+1} = \rho$. Integration of the symmetric product over the ordered simplex gives $K_0^j/j!$. The same bound works uniformly for $0 \leq z \leq \rho$. The Volterra series consequently converges uniformly there and locally uniformly in λ , solves (11), and defines its continuous endpoint. Iteration of the same estimate proves uniqueness. Positivity and the nonzero first iterate give $C(1) > 1$.

For $0 < s < \rho$, use $\cot x \leq 1/x$ for $0 < x \leq \pi/4$:

$$E(s) = \cot \frac{\rho - s}{2} \leq \frac{2}{\rho - s}, \quad K_0 \leq 2 \int_0^1 \frac{\log(1/u)}{1 - u} du = \frac{\pi^2}{3} < 4.$$

The series bound now proves all assertions. The integral identity follows by expanding $(1 - u)^{-1}$ and using monotone convergence. $\square$

In particular the exact amplitude in (2) will be

$$c = \pi C(1) = \pi A_1(\pi/2). \quad (14)$$

The endpoint is finite even though A'_1 has a logarithmic singularity there. Entire dependence on the mark does not imply that C is nonzero everywhere; its negative zeros will be important.

4 Frobenius construction uniform in complex marks

Set $t = 1 - z/\rho$ and

$$\Phi(t) = \rho t \cot(\rho t/2) = 2 + \sum_{j \geq 1} \gamma_{2j} t^{2j}, \quad \gamma_{2j} = \frac{(-1)^j 2B_{2j} \rho^{2j}}{(2j)!} = -\frac{4\zeta(2j)}{16^j}. \quad (15)$$

Here B_{2j} are Bernoulli numbers, with $B_2 = 1/6$. The radius of convergence of Φ is four. Equation (10) becomes

$$\mathcal{L}_\lambda A = 0, \quad \mathcal{L}_\lambda = (1-t) \frac{d^2}{dt^2} - \frac{d}{dt} - \lambda \frac{\Phi(t)}{t}. \quad (16)$$

The exponents at $t = 0$ are 0 and 1.

Lemma 4.1. *There are functions $H_\lambda(t)$ and $G_\lambda(t)$ analytic for $|t| < 1$, jointly holomorphic with respect to t and λ , normalized by*

$$H_\lambda(0) = 0, \quad H'_\lambda(0) = 1, \quad G_\lambda(0) = 1, \quad G'_\lambda(0) = 0,$$

such that on a fixed branch of $\log t$

$$F_\lambda = G_\lambda + 2\lambda H_\lambda \log t, \quad \mathcal{L}_\lambda F_\lambda = \mathcal{L}_\lambda H_\lambda = 0, \quad F_\lambda H'_\lambda - F'_\lambda H_\lambda = \frac{1}{1-t}. \quad (17)$$

Writing $H_\lambda = \sum_{m \geq 0} h_m(\lambda) t^m$, its explicit recurrence is

$$h_0 = 0, \quad h_1 = 1, \quad h_{m+1} = \frac{(m^2 + 2\lambda)h_m + \lambda \sum_{j=1}^{\lfloor m/2 \rfloor} \gamma_{2j} h_{m-2j}}{m(m+1)} \quad (m \geq 1). \quad (18)$$

Proof. Coefficient comparison in (16) gives (18), so every coefficient is a polynomial in λ with coefficients in $\mathbb{Q}[\rho^2]$. To see convergence uniformly on $|\lambda| \leq L$, choose any $a > 1$. The sum $\sum_{j \geq 1} |\gamma_{2j}| a^{-2j}$ is finite. For all sufficiently large m ,

$$m^2 + 2L + L \sum_{j \geq 1} |\gamma_{2j}| a^{-2j} \leq am(m+1).$$

A large enough constant D covers the preceding finitely many coefficients. Induction in (18) then gives $|h_m| \leq Da^m$. Since $a > 1$ is arbitrary, H converges jointly and locally uniformly for $|t| < 1$ and compact mark sets.

For the second solution compute

$$\mathcal{L}_\lambda(H_\lambda \log t) = \frac{2(1-t)H'_\lambda}{t} - \frac{(1-t)H_\lambda}{t^2} - \frac{H_\lambda}{t} = \frac{1}{t} + \text{an analytic function}. \quad (19)$$

The term $-2\lambda/t$ in $\mathcal{L}_\lambda G$ cancels $2\lambda/t$ from this forcing. With $g_0 = 1, g_1 = 0$, the remaining coefficient equations determine g_{m+1} for $m \geq 1$ by division only by $m(m+1)$. Their coefficients are polynomial in λ . The forcing in (19) has coefficients bounded by a polynomial in m times Da^m on each compact mark set. Choosing a slightly larger geometric factor absorbs that polynomial and gives the same majorant induction for G . Thus G too is jointly analytic on the stated domain. This constructs the pair without dividing by any parameter-dependent coefficient.

Abel's identity gives the Wronskian as a constant times $(1-t)^{-1}$. Its limit as $t \rightarrow 0$ is one, since $F_\lambda \rightarrow 1$, $H'_\lambda \rightarrow 1$, and $F'_\lambda H_\lambda \rightarrow 0$. This proves the final identity and nonsingularity of the pair for every complex λ . $\square$

Fix $t_0 \in (0, 1)$. Values of the selected solution and its derivative at t_0 are entire in λ by the Volterra construction and ordinary analytic continuation. Solve the two-by-two connection system at t_0 using (17); its determinant is never zero. Both connection coefficients are entire. Taking $t \rightarrow 0$ identifies the first with $C(\lambda)$, so

$$A_\lambda = C(\lambda)\{G_\lambda + 2\lambda H_\lambda \log t\} + D(\lambda)H_\lambda \quad (20)$$

with D entire. In particular, we never divide by $C(\lambda)$.

Define the analytic function

$$L_\lambda(t) = \rho \cot(\rho t/2) H_\lambda(t) = \sum_{j \geq 0} L_j(\lambda) t^j. \quad (21)$$

Multiplying (20) by $E(z) = \cot(\rho t/2)$ gives

$$B_\lambda(z) = \frac{2}{\rho} C(\lambda) \{t^{-1} + \lambda L_\lambda(t) \log t\} + Q_\lambda(t), \quad (22)$$

where Q is analytic in t near zero and jointly holomorphic in the mark. The term from $G_\lambda - 1$ and the term from H_λ are analytic after division by t . This argument also shows that B_λ is analytic through $z = \rho$ at every zero of C .

At $\lambda = 1$, the first coefficients are

$$H_1(t) = t + \frac{3}{2}t^2 + \frac{3}{2}t^3 + \left(\frac{11}{8} - \frac{\rho^2}{72}\right)t^4 + O(t^5),$$

$$L_0(1) = 2, \quad L_1(1) = 3, \quad L_2(1) = 3 - \frac{\rho^2}{6}, \quad (23)$$

$$L_3(1) = \frac{11}{4} - \frac{5\rho^2}{18}, \quad L_4(1) = \frac{99}{40} - \frac{3\rho^2}{10} - \frac{\rho^4}{360}.$$

Recurrences (15), (18) and (21) specify every subsequent coefficient exactly.

5 All fixed asymptotic orders with absolute uniform errors

For a nonnegative integer j let $N^{\underline{j+1}} = N(N-1)\cdots(N-j)$, and put

$$R_N(\lambda) = \frac{\rho^{N+1} P_{N+2}(\lambda)}{2(N!)^2}. \quad (24)$$

Theorem 5.1. *For every fixed integer $M \geq 1$ and compact $K \subset \mathbb{C}$,*

$$R_N(\lambda) = C(\lambda) \left\{ 1 + \lambda \sum_{j=0}^{M-1} \frac{(-1)^{j+1} j! L_j(\lambda)}{N^{\underline{j+1}}} \right\} + O_{K,M}(N^{-M-1}) \quad (25)$$

uniformly for $\lambda \in K$. The error is absolute and remains valid at zeros of C .

Proof. The meromorphic function $E(z) = \tan(\pi/4 + z/2)$ has its closest pole at ρ and its next closest pole at -3ρ . The apparent singularity of $\sec z + \tan z$ at $-\rho$ is removable. Choose $\rho < R < 3\rho$ and a slit disk $|z| < R$ cut from ρ along the positive real axis. The selected analytic solution of (10) continues there. The coefficient singularity at zero is removable for this solution. Local ODE

continuation along fixed paths and finite covers of compact transfer contours give bounds uniform on compact mark sets. Together with the convergent local Frobenius construction, this provides a fixed indented, or Δ , domain around the dominant singularity.

For $N > j$, the exact coefficient identity is

$$[z^N](1 - z/\rho)^j \log(1 - z/\rho) = \rho^{-N} \frac{(-1)^{j+1} j!}{N^{\underline{j+1}}}. \quad (26)$$

It follows either by multiplying the series for $\log(1 - z/\rho)$ by the finite binomial polynomial, or by differentiating the generalized binomial coefficient with respect to its exponent at the integer j . The pole t^{-1} contributes exactly ρ^{-N} .

To control the remainder without losing a logarithm, first retain terms of L_λ through $j = M$, one more than displayed in (25). The remaining logarithmic tail is $O_{K,M}(t^{M+1} \log t)$. Standard localized Δ -domain transfer [4, 5] gives its coefficient as $O_{K,M}(\rho^{-N} N^{-M-2} \log N)$, which is smaller than the required $\rho^{-N} N^{-M-1}$. The explicit $j = M$ term itself is $O_{K,M}(\rho^{-N} N^{-M-1})$ and may be absorbed.

For completeness, the locally analytic function Q_λ can be expanded to arbitrarily high fixed Taylor order in t . Its polynomial part has zero N th coefficient for large N ; its local high-order remainder, handled on the same contour, is smaller than the requested error. The contour away from ρ lies outside $|z| = \rho$ and contributes exponentially smaller terms. This is a localized argument: it does not say that Q_λ is globally entire or that a finite subtraction makes the full remainder exponentially small. Compact parameter bounds at each step prove the asserted uniformity. Multiplying by the normalization in (24) completes the proof. $\square$

Re-expanding each falling factorial in powers of N^{-1} gives, for every fixed order, polynomials $s_j(\lambda) \in \mathbb{Q}[\rho^2, \lambda]$ with $s_0 = 1$ and

$$R_N(\lambda) = C(\lambda) \sum_{j=0}^M \frac{s_j(\lambda)}{N^j} + O_{K,M}(N^{-M-1}).$$

The first four are

$$\begin{aligned} s_1(\lambda) &= -2\lambda, \\ s_2(\lambda) &= \lambda(2\lambda + 1), \\ s_3(\lambda) &= \frac{\lambda}{3} \{\rho^2 - (2\lambda + 1)^2\}, \\ s_4(\lambda) &= \frac{\lambda}{6} (4\lambda^3 + 4\lambda^2 + \lambda - 7\lambda\rho^2 + 3\rho^2). \end{aligned} \quad (27)$$

In this notation the bracket is multiplied by C and followed by an absolute error; it is not a relative-error statement at a zero of C . At $\lambda = 0$, $A_0 = 1$ and $B_0 = E$ exactly, and all algebraic correction polynomials vanish.

Corollary 5.2. *For any fixed $M \geq 1$,*

$$d_{N+2} = 2C(1)\rho^{-N-1}(N!)^2 \left\{ 1 + \sum_{j=0}^{M-1} \frac{(-1)^{j+1} j! L_j(1)}{N^{\underline{j+1}}} + O_M(N^{-M-1}) \right\}. \quad (28)$$

The first six ordinary inverse-power corrections are

$$1 - \frac{2}{N} + \frac{3}{N^2} + \frac{-3 + \rho^2/3}{N^3} + \frac{3/2 - 2\rho^2/3}{N^4} + \frac{3/5 - 7\rho^2/15 + \rho^4/15}{N^5} + \frac{-6/5 + 13\rho^2/30 + 19\rho^4/45}{N^6} + O(N^{-7}). \quad (29)$$

In particular (2) holds with $c = \pi C(1)$.

Proof. Set $\lambda = 1$ and use $C(1) > 0$ to convert the absolute error to a relative error. The displayed coefficients follow by formal re-expansion of the finite rational sum. Finally Stirling's formula gives

$$2C(1)\rho^{1-n}\Gamma(n-1)^2 \sim 4\pi C(1)\rho^{1-n}n^{2n-3}e^{-2n},$$

which is (2) with (14). □

There are no $\log N$ factors in this single dominant algebraic expansion. The local solution has just one logarithm, and its integer-power logarithmic coefficients are the exact rational expressions (26). This does not address all farther singularities or a full exponential transseries.

6 An exact Wronskian formula and a certified amplitude

Write $A = A_1$, $H = H_1$. Both the endpoint definition and the following geometrically convergent formula specify the same constant.

Proposition 6.1. For $0 < z < \rho$, with $t = 1 - z/\rho$,

$$C(1) = z \left\{ \frac{A(z)H'(t)}{\rho} + A'(z)H(t) \right\}. \quad (30)$$

In particular

$$C(1) = \frac{A(\rho/2)H'(1/2) + \rho A'(\rho/2)H(1/2)}{2}. \quad (31)$$

Proof. The z -equation is $A'' + A'/z - EA/z = 0$. Abel's identity says that z times the Wronskian of two solutions is constant. For the solutions $A(z)$ and $H(1 - z/\rho)$, the limit at $z \uparrow \rho$ is $-C(1)$: the first derivative of the latter tends to $-1/\rho$, while its value tends to zero and A' grows only logarithmically. This gives (30), with its sign as displayed, and then (31). □

We give explicit tails so that the decimal enclosure is a finite certificate rather than just a high-precision observation. From (7) and positivity,

$$q_n = p_n \rho^n \geq 0, \quad \sum_{n \geq 0} q_n = C(1) < 200.$$

Thus, truncating through degree $m \geq 1$ at $z = \rho/2$,

$$\begin{aligned} \epsilon_A &= 200 \cdot 2^{-m-1}, & |A - A_m| &\leq \epsilon_A, \\ \epsilon_D &= 200(m+1)2^{-m}, & |\rho A' - \rho A'_m| &\leq \epsilon_D. \end{aligned} \quad (32)$$

The derivative bound uses the decrease of $n2^{1-n}$ for $n \geq 2$. The full values satisfy $|A| \leq 200$ and $|\rho A'| \leq 200$ because $n2^{1-n} \leq 1$ for $n \geq 1$.

A convenient majorant for H is

$$|h_n(1)| \leq \frac{2}{3} \left(\frac{3}{2}\right)^n \quad (n \geq 1). \quad (33)$$

Indeed, with $a = 3/2$ and (15),

$$\sum_{j \geq 1} |\gamma_{2j}| a^{-2j} = 4 \sum_{j \geq 1} \frac{\zeta(2j)}{36^j} < 8 \sum_{j \geq 1} 36^{-j} = \frac{8}{35}.$$

The $m = 1$ step of (18) is equality in (33). For $m \geq 2$ the induction follows from

$$m^2 + 2 + \frac{8}{35} \leq \frac{3}{2} m(m+1).$$

This proves $|H(1/2)| \leq 2$, $|H'(1/2)| \leq 16$, and the geometric tails

$$\begin{aligned} \epsilon_H &= \frac{8}{3} \left(\frac{3}{4}\right)^{m+1}, & |H - H_m| &\leq \epsilon_H, \\ \epsilon_T &= \frac{16}{3} (m+4) \left(\frac{3}{4}\right)^{m+1}, & |H' - H'_m| &\leq \epsilon_T. \end{aligned} \quad (34)$$

Consequently the midpoint product error in (31) is at most

$$\delta_m = \frac{1}{2} (16\epsilon_A + 200\epsilon_T + \epsilon_A\epsilon_T + 2\epsilon_D + 200\epsilon_H + \epsilon_D\epsilon_H). \quad (35)$$

For example, one may write each partial sum as its full value plus its tail and bound the resulting cross terms. The estimate is deliberately conservative.

The implementation takes $m = 420$. Exact rational arithmetic verifies $\delta_{420} < 10^{-44}$; the value is approximately $5.7 \cdot 10^{-48}$. The finite recurrences and products use outward interval arithmetic at 90 decimal places, with interval-valued π and enclosed rational inputs. Adding $[-\delta_{420}, \delta_{420}]$ before multiplying by the interval for π gives the following shortened outward endpoints:

$$\begin{aligned} 25.5745196289574675215372323127353368945234275278994 &< c, \\ c < 25.5745196289574675215372323127353368945234275279353. \end{aligned} \quad (36)$$

These inequalities are the certified numerical claim. The certificate depends on the usual correctness of the interval implementation; it is not a proof-assistant verification of that software. The exact core separately checks the rational tail inequalities.

For orientation, ordinary multiprecision evaluation gives

$$C(1) = 8.1406224322985719224911291170229833094279980423378659572 \dots$$

The companion diagnostics evaluate the Wronskian at $z/\rho = 0.4, 0.5, 0.6$ and compare long strings. Agreement is useful for detecting mistakes, but those strings are not additional interval-certified digits. Likewise, the amplitude enclosure does not provide effective finite- N constants for (25).

[write] Added 7 October 2026. With its own code (exact rational Taylor coefficients from (7), the recurrence (18) with $\gamma_{2j} = -4\zeta(2j)/16^j$, $m = 420$, 70 digits; floating, not interval) the write evaluated (31): $C(1) = 8.14062243229857192249112911702298330942799804233786595723 \dots$ and $\pi C(1) = 25.57451962895746752153723231273533689452342752791735 \dots$. The latter lies inside (36); the source's printed $C(1)$ and the OEIS decimal of c are truncations of these values.

7 Logarithmic carriers and discrete inversion

Set

$$d = \frac{1}{e\sqrt{\rho}}, \quad C_* = \log(2\pi c). \quad (37)$$

Taking the logarithm of the scalar expansion and applying Stirling with $N = n - 2$ yields

$$\begin{aligned} \log d_n &= 2n \log(dn) - 3 \log n + C_* + \frac{1}{6n} - \frac{2}{n^2} \\ &\quad + \frac{\rho^2/3 - 541/180}{n^3} + O(n^{-4}). \end{aligned} \quad (38)$$

To check the change of indexing directly, use $\Gamma(n-1) = \Gamma(n+1)/(n(n-1))$ and

$$2 \log \Gamma(n+1) = (2n+1) \log n - 2n + \log(2\pi) + \frac{1}{6n} - \frac{1}{180n^3} + O(n^{-5}),$$

then expand the finite correction bracket in powers of $1/(n-2)$. In particular the $1/n$ and $1/n^2$ terms in (38) are not those of (29).

For $L = \log X$ and sufficiently large X , define

$$x_0 = \frac{L}{2W_0(dL/2)}, \quad x_1 = x_0 + \frac{3 \log x_0 - C_*}{2 \log(x_0/\sqrt{\rho})}, \quad (39)$$

where W_0 is the positive real branch of Lambert W . The leading function $f(x) = 2x \log(dx)$ has $f(x_0) = L$ and $f'(x) = 2 \log(x/\sqrt{\rho})$. The correction in (39) cancels the residual $-3 \log x_0 + C_*$. It tends to $3/2$ and is bounded. Taylor's theorem gives a remaining residual $O(1/x_0)$ in any sufficiently accurate logarithmic carrier from (38). Its derivative is asymptotic to $2 \log x$, so the corresponding smooth inverse is

$$x = x_1 + O\left(\frac{1}{x_0 \log x_0}\right). \quad (40)$$

This describes an inverse of a chosen asymptotic carrier, not a canonical interpolation of the integer sequence.

For an unambiguous integer statement fix $M \geq 1$ and define the real carrier

$$F_M(x) = 2C(1)\rho^{1-x}\Gamma(x-1)^2 \left\{ 1 + \sum_{j=0}^{M-1} \frac{(-1)^{j+1} j! L_j(1)}{(x-2)^{j+1}} \right\}. \quad (41)$$

It is positive for all sufficiently large real x , and its logarithmic derivative is asymptotic to $2 \log x$. Let $\xi_M(X)$ denote its unique eventual inverse.

Theorem 7.1. *There are constants $K_M > 0$ and X_M such that, for $X \geq X_M$, writing $\xi = \xi_M(X)$,*

$$\left\lceil \xi - \frac{K_M \xi^{-M-1}}{\log \xi} \right\rceil \leq \min\{n \geq 3 : d_n \geq X\} \leq \left\lceil \xi + \frac{K_M \xi^{-M-1}}{\log \xi} \right\rceil. \quad (42)$$

Proof. Equation (28) implies $\log d_n = \log F_M(n) + O_M(n^{-M-1})$. The exact sequence is strictly increasing from $n = 2$ onward: for $n \geq 3$, the initial boundary term and the $k = 2$ increment give $d_n \geq d_{n-1} + (n-2)T(n-1, n-2) = (n-1)d_{n-1} > d_{n-1}$.

For x near ξ , a lower bound $c_0 \log \xi$ for $(\log F_M)'(x)$ converts a displacement of $K_M \xi^{-M-1}/\log \xi$ into a logarithmic change exceeding the remainder bound when K_M is sufficiently large. Thus

every integer below the left real endpoint has $d_n < X$, while every integer at or above the right endpoint has $d_n \geq X$. The estimates hold first on a fixed proportional neighborhood of ξ ; outside that neighborhood monotonicity and the same boundary comparisons suffice. Taking ceilings yields (42). Enlarge the eventual constants to cover the finite onset of the comparison. $\square$

The envelope tends to zero and eventually the two integer bounds differ by at most one. If the inverse is farther from an integer than the envelope, both bounds coincide. Close to an integer boundary an exact sequence evaluation is still necessary. Neither K_M nor X_M has been computed here as an effective finite-input certificate. Taking $\lceil x_1 \rceil$, or even $\lceil \xi_M \rceil$, is not asserted to work for every large threshold. Higher fixed-order refinements can be obtained from (41) by Newton iteration with controlled Taylor remainders.

Remark 7.2 ([write] The inverses against the transseries volume, 7 October 2026). The volume is `Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`; its symbols carry the subscript “vol”. Statement by statement:

- (a) *Instance.* $N(X) = \min\{n \geq 3 : d_n \geq X\}$ is the integer staircase of `p0:def:three-inverses(1)` with $A_n = d_n$ and $n_{1,\text{vol}} = 3$ (d_n is strictly increasing from $n = 2$, as the proof of Theorem 7.1 shows), so `p0:thm:staircase(1)` gives $N(X) = \lceil \nu_{\mathcal{A}}(X) \rceil$ for every admissible interpolation $\mathcal{A}_{\text{vol}}$ of (d_n) .
- (b) *Analogue.* Theorem 7.1, as stated and proved, is a direct two-ceiling bracket at the integers about the root ξ_M of the carrier (41), which does not interpolate the sequence: an analogue of `p0:thm:staircase(2)`, not an instance. It can be routed through `p0:thm:staircase(1)` as follows; this is the same comparison rearranged, not an independent proof. Let $E_n = \log d_n - \log F_M(n)$, so $|E_n| \leq C_M n^{-M-1}$ by (28), let $\tilde{E}$ be its piecewise-linear interpolation on $[n_2, \infty)$ and $\mathcal{A}_{\text{vol}} = F_M e^{\tilde{E}}$. On each $[n, n+1]$, $|\tilde{E}'| \leq |E_n| + |E_{n+1}| \leq 2C_M n^{-M-1}$ while $(\log F_M)' \geq c_0 \log x$, so for n_2 large $\mathcal{A}_{\text{vol}}$ is continuous, strictly increasing and equal to d_n at the integers $n \geq n_2$: it is admissible (with $n_{1,\text{vol}} = n_2$). Its inverse ν satisfies $\log F_M(\nu) = \log F_M(\xi) - \tilde{E}(\nu)$ with $|\tilde{E}(\nu)| \leq C_M \lfloor \nu \rfloor^{-M-1}$; hence $\nu/\xi \rightarrow 1$, and the mean value theorem gives $|\nu - \xi| \leq K_M \xi^{-M-1}/\log \xi$ for a suitable K_M and large X . For $X > d_{n_2}$ the staircase from n_2 equals $N(X)$, so `p0:thm:staircase(1)` and the monotonicity of the ceiling give (42).
- (c) *Instance.* The Lambert start x_0 of (39) solves $2x \log(dx) = L$, which is `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 2$ and $d_{\text{vol}} = 2 \log d = -(2 + \log \rho)$: its solution $X_{\text{vol}} = \exp(v - d_{\text{vol}}/2) = e^v/d$ with $v = W_0(dL/2)$ equals $L/(2v)$ because $ve^v = dL/2$. The correction x_1 is one Newton step, not a statement of the volume.
- (d) *Instance, and analogue.* In Corollary 10.4, $n(y)$ is the staircase of the strictly increasing sequence $A_m = -\log q_m$ ($A_0 = 0$; every mass is positive) at the level $T = \log(1/y)$, with $n_{1,\text{vol}} = 0$: an instance of `p0:def:three-inverses(1)`, so `p0:thm:staircase(1)` applies. The bracket (70) is again an analogue of `p0:thm:staircase(2)`; the argument of (b) applies with $|A_m - H(m)| = O(m^{-1/2})$ in place of $C_M n^{-M-1}$. Its start x_0 of (71) is `p0:prop:factorial-core` with $\kappa_{\text{vol}} = 2$, $d_{\text{vol}} = 2 \log(2/(eA))$ ($v = W_0(T/(eA))$, $X_{\text{vol}} = T/(2v)$).

No novelty is claimed for any of these inversions. The staircase arithmetic is formalized generically (for an arbitrary strictly monotone function) as `Fabius.staircase_ceil` in `Analysis/FabiusFunction/Lean/FabiusFunction/StaircaseInversion.lean`; nothing about d_n is formalized.

8 A Green operator despite the infinite measure

The endpoint connection has a useful spectral interpretation. Put

$$\mu(ds) = E(s) ds \quad (0 < s < \rho), \quad \mathcal{H} = L^2((0, \rho), \mu), \quad \mathcal{G} = L^2((0, \rho), dx/x).$$

The measure μ has infinite total mass near ρ , but is σ -finite. Define

$$(Sf)(x) = \int_0^x f(s) \mu(ds), \quad k(s, t) = \log \frac{\rho}{\max(s, t)}. \quad (43)$$

For each $x < \rho$, the integral defining Sf exists by Cauchy–Schwarz, since $\mu((0, x)) < \infty$.

Theorem 8.1. *The operator $S : \mathcal{H} \rightarrow \mathcal{G}$ is Hilbert–Schmidt, with $\|S\|_{\text{HS}}^2 = K_0$. The operator $K = S^*S$ is positive, injective and trace class, with kernel k and an infinite sequence of positive eigenvalues κ_j satisfying $\sum_j \kappa_j = K_0$. Moreover*

$$C(\lambda) = \det(I + \lambda K) = \prod_{j \geq 1} (1 + \lambda \kappa_j) \quad (44)$$

locally uniformly for every complex λ .

Proof. The kernel of S between the two L^2 spaces is $\mathbf{1}_{\{s < x\}}$. Tonelli’s theorem gives

$$\|S\|_{\text{HS}}^2 = \int_0^\rho \int_0^x E(s) ds \frac{dx}{x} = \int_0^\rho E(s) \log \frac{\rho}{s} ds = K_0 < \infty.$$

It follows that $K = S^*S$ is positive trace class. The kernel product gives

$$\int_{\max(s, t)}^\rho \frac{dx}{x} = k(s, t), \quad \text{Tr } K = \|S\|_{\text{HS}}^2 = K_0.$$

If $Sf = 0$ almost everywhere, its locally absolutely continuous representative is identically zero on $(0, \rho)$. Its derivative is $E(s)f(s)$ almost everywhere, so $f = 0$. Thus S and K are injective. The space $\mathcal{H}$ is infinite dimensional, and a finite-rank compact operator cannot be injective there. The spectral theorem therefore gives infinitely many strictly positive eigenvalues with the stated sum.

For ordered points $0 < s_1 < \dots < s_j < \rho$, the determinant of the kernel matrix is

$$\det[k(s_i, s_\ell)]_{i, \ell=1}^j = \log \frac{\rho}{s_j} \prod_{i=1}^{j-1} \log \frac{s_{i+1}}{s_i}. \quad (45)$$

To see this, write $u_i = \log(\rho/s_i)$ and reverse both row and column order. The kernel matrix becomes $[\min(u_i, u_\ell)]$ with increasing u_i . It factors as a unit lower-triangular matrix, the diagonal matrix of successive increments, and the transpose of the first matrix. Multiplying the increments proves (45). By symmetry, its $1/j!$ integral over the full product space is the ordered integral. That integral is precisely $(V^j \mathbf{1})(\rho)$.

We justify the Fredholm expansion rather than assuming finite total measure. Let P_ϵ be multiplication by $\mathbf{1}_{[\epsilon, \rho-\epsilon]}$. On that compact interval the continuous kernel has the ordinary Fredholm series. Dominated convergence in the Hilbert–Schmidt norm gives $\|S - SP_\epsilon\|_{\text{HS}} \rightarrow 0$. The decomposition

$$K - P_\epsilon K P_\epsilon = (S - SP_\epsilon)^* S + (SP_\epsilon)^* (S - SP_\epsilon)$$

and the Hilbert–Schmidt product inequality imply trace-norm convergence. Hence the Fredholm determinants converge locally uniformly in λ .

The ordered coefficients converge as well by dominated convergence. They are bounded by $K_0^j/j!$, exactly as in Theorem 3.1, which permits locally uniform passage through the series in λ . Thus the limit of the Fredholm series equals the endpoint Volterra series. This proves the first equality in (44). The standard spectral product for a trace-class operator gives the second; alternatively it follows from finite spectral truncations and the same trace-norm continuity. $\square$

This is a Green-kernel determinant and a classical min-covariance factorization applied to the particular weight $E(s)$. The infinite total mass of μ is harmless because the relevant two-variable kernel has the required Hilbert–Schmidt factorization. If point-process language is used, K is an L -ensemble or Green operator; the correlation operator is $K(I + K)^{-1}$, with eigenvalues in $(0, 1)$. These two operators must not be identified.

9 The limiting law and every fixed probability correction

Let

$$G(\lambda) = \frac{C(\lambda)}{C(1)}, \quad p_j = \frac{\kappa_j}{1 + \kappa_j}, \quad \mu_J = \sum_j p_j, \quad v_J = \sum_j p_j(1 - p_j). \quad (46)$$

Theorem 9.1. *The limiting probability generating function is*

$$G(\lambda) = \prod_{j \geq 1} (1 - p_j + p_j \lambda). \quad (47)$$

It is the law of $J = \sum_{j \geq 1} Y_j$, where the Y_j are independent Bernoulli variables with success probabilities p_j . This sum is almost surely finite, has every fixed exponential moment, and assigns positive probability to every nonnegative integer. Its mean and variance are μ_J and v_J , with $0 < v_J < \mu_J < \infty$, and $\mathbb{P}(J = 0) = 1/C(1)$.

Proof. Equation (47) is (44) divided by the positive scalar $C(1)$. Since $\sum_j p_j \leq \sum_j \kappa_j = K_0 < \infty$, the independent sum is finite almost surely, for example by the first Borel–Cantelli lemma. Its finite partial-sum PGFs converge to the displayed product, locally uniformly. For every real $a > 0$,

$$\mathbb{E}e^{aJ} = \prod_j \{1 + p_j(e^a - 1)\} \leq \exp\{(e^a - 1) \sum_j p_j\} < \infty.$$

Termwise expectations and variances are therefore justified. Every p_j lies strictly between zero and one. The sum of p_j^2 is strictly positive, so $v_J = \mu_J - \sum_j p_j^2 < \mu_J$. Fixing successes for any chosen k indices and failures for all remaining indices has positive probability: the product of failure probabilities is positive because $\sum p_j < \infty$. This gives positive mass at every k . Taking $\lambda = 0$ gives the zero mass. $\square$

This spectral Bernoulli product is a standard principle; compare the point-count theorem in HKPV [6, Theorem 7]. Here it follows directly from the determinant and does not imply that the finite polynomials P_n themselves factor into Bernoulli factors.

For fixed-order corrections define $Q_0(\lambda) = 1$ by formal division

$$\sum_{j \geq 0} Q_j(\lambda) u^j = \frac{\sum_{j \geq 0} s_j(\lambda) u^j}{\sum_{j \geq 0} s_j(1) u^j}, \quad Q_m(\lambda) = s_m(\lambda) - \sum_{i=1}^m s_i(1) Q_{m-i}(\lambda). \quad (48)$$

These are formal series, used only to any fixed finite degree. The first terms are

$$\begin{aligned} Q_1(\lambda) &= 2(1 - \lambda), \\ Q_2(\lambda) &= (\lambda - 1)(2\lambda - 1), \\ Q_3(\lambda) &= \frac{\lambda - 1}{3}(-4\lambda^2 + 4\lambda + \rho^2 + 3). \end{aligned} \tag{49}$$

In particular $Q_j(1) = 0$ for $j > 0$.

Theorem 9.2. *For every fixed $M \geq 0$ and compact $K \subset \mathbb{C}$,*

$$G_N(\lambda) = G(\lambda) \sum_{j=0}^M \frac{Q_j(\lambda)}{N^j} + O_{K,M}(N^{-M-1}) \tag{50}$$

uniformly on K . Define signed coefficients $\nu_j(k) = [\lambda^k]G(\lambda)Q_j(\lambda)$. For every $w > 0$,

$$\sum_{k \geq 0} w^k \left| \mathbb{P}(J_N = k) - \sum_{j=0}^M \frac{\nu_j(k)}{N^j} \right| = O_{w,M}(N^{-M-1}). \tag{51}$$

Consequently $J_N \rightarrow J$ in total variation and with every fixed exponential moment, and all fixed moments have corresponding expansions.

Proof. The exact ratio is $G_N(\lambda) = R_N(\lambda)/R_N(1)$. The denominator tends to $C(1) > 0$. Divide its scalar finite expansion into the absolute compact-uniform numerator expansion of Theorem 5.1. This proves (50), even at zeros of $C(\lambda)$, because the only division by an endpoint function is at $\lambda = 1$.

For (51), choose $R > w$ and apply Cauchy's estimate to the entire difference in (50) on $|\lambda| = R$. Its k th coefficient is bounded in absolute value by $C_{R,M}N^{-M-1}R^{-k}$. Summing the geometric series gives the explicit bound

$$\frac{C_{R,M}}{1 - w/R} N^{-M-1}.$$

At $w = 1$ this controls the full absolute mass difference (twice the usual total variation distance at order zero). At larger w , polynomial weights and fixed exponential weights are dominated, so moments follow. Alternatively derivatives near $\lambda = 1$ follow directly from Cauchy estimates. $\square$

For practical use, the first signed corrections may be written without coefficient extraction. Let $p(k) = \mathbb{P}(J = k)$, with $p(k) = 0$ for $k < 0$. Then

$$\nu_1(k) = 2p(k) - 2p(k-1), \quad \nu_2(k) = p(k) - 3p(k-1) + 2p(k-2).$$

They have total mass zero, as do all higher corrections.

The logarithm of the correction factor in (50), expanded near $\lambda = 1$, is

$$\frac{2(1-\lambda)}{N} + \frac{\lambda-1}{N^2} + \frac{2\lambda^2 + \lambda\rho^2 - \lambda - \rho^2 - 1}{3N^3} + O(N^{-4}). \tag{52}$$

Using $\lambda d/d\lambda$ for ordinary cumulants gives

$$\mathbb{E}J_N = \mu_J - \frac{2}{N} + \frac{1}{N^2} + \frac{\rho^2 + 3}{3N^3} + O(N^{-4}), \tag{53}$$

$$\text{Var } J_N = v_J - \frac{2}{N} + \frac{1}{N^2} + \frac{\rho^2 + 7}{3N^3} + O(N^{-4}). \tag{54}$$

The constants μ_J and v_J have the exact convergent spectral definitions (46); no unverified numerical eigenvalue computation is needed.

10 Growth for a large positive boundary mark

The preceding coefficient asymptotics keep the mark in a fixed compact set. We now study the endpoint function itself in a different limit, using the exact differential equation. Define the finite optical length

$$\mathcal{A} = \int_0^\rho \sqrt{\frac{E(z)}{z}} dz. \quad (55)$$

Both endpoint singularities are integrable. Numerically $\mathcal{A} \approx 4.2586174557$, an ordinary quadrature value rather than an interval certificate.

Theorem 10.1. *As the real parameter $\lambda \rightarrow +\infty$,*

$$C(\lambda) = \frac{\exp(\mathcal{A}\sqrt{\lambda})}{2\sqrt{2}\pi\sqrt{\lambda}} \{1 + O(\lambda^{-1/2})\}. \quad (56)$$

The implied constant is an existence constant. This statement gives neither complex-sector uniformity nor a simultaneous limit with $N \rightarrow \infty$.

We first record the endpoint comparison needed for a proof. The modified Bessel functions I_ν and K_ν have positive values on the positive real axis for $\nu = 0, 1$, and their Wronskian, endpoint limits and large-argument expansions are classical [7].

Lemma 10.2. *Fix $a > 0$, $\nu \in \{0, 1\}$, and a bounded continuous real function q on $[0, a]$. In*

$$u'' = \left(k^2 + \frac{\nu^2 - 1/4}{x^2} + q(x) \right) u, \quad k \rightarrow +\infty,$$

select the solution regular at zero with the same leading coefficient as $c_k \sqrt{x} I_\nu(kx)$, where $c_k \neq 0$ may depend on k . For the endpoint-analytic coefficients used below, this solution is unique. At the fixed point a ,

$$u(a) = c_k \sqrt{a} I_\nu(ka) \{1 + O(k^{-1})\}, \quad u'(a) = c_k \frac{d}{dx}(\sqrt{x} I_\nu(kx)) \Big|_{x=a} \{1 + O(k^{-1})\}. \quad (57)$$

Proof. Put $f(x) = \sqrt{x} I_\nu(kx)$ and $h(x) = \sqrt{x} K_\nu(kx)$. Their Wronskian is $fh' - f'h = -1$, so f/h is increasing. The regular Green kernel

$$\mathcal{G}_k(x, t) = f(x)h(t) - h(x)f(t) \quad (0 < t < x)$$

is nonnegative and has diagonal x -derivative one. Let $B_\nu = \sup_{y>0} y I_\nu(y) K_\nu(y) < \infty$; finiteness follows from the classical zero and infinity limits and continuity on intervening compact intervals. Consequently

$$0 \leq \frac{\mathcal{G}_k(x, t)f(t)}{f(x)} \leq t I_\nu(kt) K_\nu(kt) \leq \frac{B_\nu}{k}. \quad (58)$$

The equation $u = c_k f + \int_0^x \mathcal{G}_k(x, t) q(t) u(t) dt$ has a unique normalized solution $r = u/(c_k f)$. Volterra iteration and Gronwall give, with $M = \|q\|_\infty$,

$$|r(x)| \leq e^{MB_\nu x/k}, \quad |r(x) - 1| \leq e^{MB_\nu x/k} - 1.$$

This proves the value estimate. At zero the constructed solution has the specified regular leading coefficient. Reduction of order gives an excluded $\sqrt{x} \log x$ solution for $\nu = 0$ and an excluded $x^{-1/2}$

solution for $\nu = 1$, so it is the selected endpoint solution. In our application the coefficients have even analytic endpoint expansions, also giving this uniqueness directly by regular-singular theory.

Derivative control is separate. At fixed a , the Bessel derivative expansions give $f'(a) > 0$, $h'(a) < 0$ for large k and

$$\frac{|h'(a)|}{f'(a)} \leq D \frac{K_\nu(ka)}{I_\nu(ka)}$$

with D independent of k . Since K_ν/I_ν is decreasing, for $0 < t < a$,

$$\frac{|\partial_x \mathcal{G}_k(a, t)| f(t)}{f'(a)} \leq t I_\nu(kt) K_\nu(kt) + D \frac{K_\nu(ka)}{I_\nu(ka)} t I_\nu(kt)^2 \leq \frac{(1+D)B_\nu}{k}.$$

This supplies an integrable majorant for differentiating the Volterra equation at a ; the moving endpoint contributes zero because $\mathcal{G}_k(a, a) = 0$. Combining the bound with the estimate for r proves the derivative assertion in (57). No uncontrolled error term was differentiated. $\square$

Proof of Theorem 10.1. Put $k = \sqrt{\lambda}$ and use the Liouville variables

$$x = \int_0^z \sqrt{E(s)/s} ds, \quad g = (zE(z))^{1/4}, \quad \psi = gA_\lambda. \quad (59)$$

A direct substitution in $(zA'_\lambda)' = k^2 EA_\lambda$ gives

$$\psi_{xx} = (k^2 + Q)\psi, \quad Q = \frac{g_{xx}}{g}, \quad 0 < x < \mathcal{A}.$$

The derivatives on g here are with respect to x . At the left endpoint, analytic inversion in $\sqrt{z}$ gives

$$g = \sqrt{x/2} \{1 + x^2/24 + O(x^4)\}, \quad Q = -\frac{1}{4x^2} + \frac{1}{6} + O(x^2). \quad (60)$$

Indeed $x = 2\sqrt{z}$ times an analytic function of z with constant term one. At the right endpoint write $\delta = \rho - z$ and $v = \mathcal{A} - x$. Since $E(z) = \cot(\delta/2)$,

$$v = 2\sqrt{2\delta/\rho} \{1 + O(\delta)\}, \quad g = 2v^{-1/2} h(v^2), \quad h(0) = 1, \quad Q = \frac{3}{4v^2} + O(v^2), \quad (61)$$

where h is analytic. Analyticity justifies differentiated expansions. Thus after subtracting the displayed inverse-square term, the left and right potentials are bounded on any fixed respective interval up to an interior matching point.

The selected left solution has $\psi_A \sim \sqrt{x/2}$. The right solution $H_\lambda(t) = t + O(t^2)$, with $t = 1 - z/\rho = v^2/8 + O(v^4)$, gives $\psi_H = gH_\lambda \sim v^{3/2}/4$. Their correctly normalized Bessel models are

$$U_\ell(x) = \sqrt{x/2} I_0(kx), \quad U_r(v) = \frac{\sqrt{v} I_1(kv)}{2k}.$$

Fix $a \in (0, \mathcal{A})$ and write $b = \mathcal{A} - a$. Apply Lemma 10.2 at both endpoints. The standard positive-axis Bessel expansions then give

$$\begin{aligned} \psi_A(a) &= \frac{e^{ka}}{\sqrt{4\pi k}} \{1 + O(k^{-1})\}, & \psi_{A,x}(a) &= \frac{ke^{ka}}{\sqrt{4\pi k}} \{1 + O(k^{-1})\}, \\ \psi_H(a) &= \frac{e^{kb}}{2k\sqrt{2\pi k}} \{1 + O(k^{-1})\}, & \psi_{H,x}(a) &= -\frac{ke^{kb}}{2k\sqrt{2\pi k}} \{1 + O(k^{-1})\}. \end{aligned}$$

The last minus sign is from $v = \mathcal{A} - x$. The exact Wronskian identity is

$$W_x(\psi_A, \psi_H) = g^2 \frac{dz}{dx} W_z(A_\lambda, H_\lambda) = z W_z(A_\lambda, H_\lambda) = -C(\lambda).$$

The constant follows either from Abel's identity or by taking $z \uparrow \rho$: $H_\lambda \sim (\rho - z)/\rho$, $(H_\lambda)_z \rightarrow -1/\rho$, and $A'_\lambda = O(|\log(\rho - z)|)$ for each fixed λ . Thus the product with H_λ vanishes and the other product tends to $-C(\lambda)$. At the fixed interior point the two Wronskian terms have the same negative leading sign. Their sum is

$$-C(\lambda) = -\frac{e^{k\mathcal{A}}}{2\sqrt{2}\pi k} \{1 + O(k^{-1})\},$$

which proves (56). $\square$

For stable quadrature, the substitution $z = \rho \sin^2 \theta$ removes both endpoint singularities:

$$\mathcal{A} = 2\sqrt{2} \int_0^{\pi/2} \sqrt{u \cot u} d\theta, \quad u = \frac{\pi}{4} \cos^2 \theta,$$

with $u \cot u = 1$ at $u = 0$. The optional numerical check uses this smooth formula. The proof uses only its exact integral. Liouville transformation, Bessel comparison and Wronskian matching are classical methods; no new general asymptotic method is asserted. In particular, (56) does not license inserting a growing λ into Theorem 5.1 or extracting finite-size rare-tail probabilities without further uniform analysis.

[write] **Added 7 October 2026: a diagnostic.** The write evaluated $C(\lambda)$ for real λ by the marked analogue of (30) at $z = \rho/2$, $C(\lambda) = z\{A_\lambda(z)H'_\lambda(t)/\rho + A'_\lambda(z)H_\lambda(t)\}$ (900 Taylor and Frobenius terms, 60 digits, floating). The ratio of $C(\lambda)$ to the right-hand side of (56) without its error factor is 1.005513, 1.002776, 1.001390, 1.000695 at $\lambda = 25, 100, 400, 1600$, and $\sqrt{\lambda}$ times its excess over one is 0.02756, 0.02776, 0.02780, 0.02781: consistent with the stated $O(\lambda^{-1/2})$ error, and suggesting (not proving) a first correction $c_1 \lambda^{-1/2}$ with $c_1 \approx 0.0278$; see item 4 of Section 13.

10.1 Rare counts in the fixed limiting law

The determinant product supplies enough structure to extract a far-tail theorem from the real-positive estimate alone. This concerns the single limiting variable J of Theorem 9.1. It is not a joint limit for the finite variables J_N .

Theorem 10.3. Put $p(m) = \mathbb{P}(J = m)$ and

$$K_{\text{tail}} = \frac{\mathcal{A}}{4\sqrt{2}\pi^{3/2}C(1)}. \quad (62)$$

As the integer $m \rightarrow \infty$,

$$p(m) = \frac{\mathcal{A}^{2m+1}}{\sqrt{2}\pi C(1)(2m+1)!} \{1 + O(m^{-1/2})\} \quad (63)$$

$$= K_{\text{tail}} m^{-3/2} \left(\frac{e\mathcal{A}}{2m}\right)^{2m} \{1 + O(m^{-1/2})\}. \quad (64)$$

The upper tail satisfies

$$\frac{\mathbb{P}(J \geq m)}{p(m)} = 1 + \frac{\mathcal{A}^2}{4m^2} + o(m^{-2}). \quad (65)$$

All error constants and onsets are non-effective here. The factorial comparison is asymptotic; it does not identify an exact conditioned-Poisson law.

Proof. For each $r > 0$, exponential tilting of the positive product gives independent Bernoulli variables with parameters

$$p_j(r) = \frac{r\kappa_j}{1 + r\kappa_j}, \quad M(r) = \sum_j p_j(r), \quad V(r) = \sum_j p_j(r)(1 - p_j(r)) = rM'(r).$$

Their mean is finite because $\sum p_j(r) \leq rK_0$. The sums and their derivatives converge locally uniformly in r . Also $M'(r) = \sum_j \kappa_j / (1 + r\kappa_j)^2$ is positive and decreasing.

We first obtain derivative estimates without differentiating an asymptotic error. Set $s = \log r$, $k = e^{s/2}$ and $f(s) = \log C(e^s)$. Theorem 10.1 gives

$$f(s) = \mathcal{A}k - s/2 - \log(2\sqrt{2}\pi) + O(k^{-1}), \quad f' = M(e^s), \quad f'' = V(e^s) \geq 0.$$

Convexity bounds $f'(s)$ between its forward and backward secants at step $h = k^{-1}$. The leading secant error is $O(kh) = O(1)$, while division of the two remainder values by h gives $O(1/(kh)) = O(1)$. Hence

$$M(r) = \frac{\mathcal{A}}{2}\sqrt{r} + O(1). \quad (66)$$

For $0 < \delta < 1$, monotonicity of M' yields

$$\frac{M(r(1 + \delta)) - M(r)}{\delta} \leq V(r) \leq \frac{M(r) - M(r(1 - \delta))}{\delta}.$$

Using (66) and $\delta = r^{-1/4}$ gives

$$V(r) = \frac{\mathcal{A}}{4}\sqrt{r} + O(r^{1/4}). \quad (67)$$

The strictly increasing continuous M ranges from zero to infinity. Thus there is a unique $r_m > 0$ with $M(r_m) = m$, and, writing $k_m = \sqrt{r_m}$,

$$k_m = \frac{2m}{\mathcal{A}} + O(1), \quad V(r_m) = \frac{m}{2} + O(\sqrt{m}). \quad (68)$$

We need a uniform elementary local estimate. For any finite or countable independent Bernoulli sum S with finite mean M , variance $V \rightarrow \infty$, and integer mean M ,

$$\mathbb{P}(S = M) = \frac{1}{\sqrt{2\pi V}} + O(V^{-1}) = \frac{1 + O(V^{-1/2})}{\sqrt{2\pi V}}, \quad (69)$$

with an absolute error constant. Here is a proof that also covers the infinite family. For its centered characteristic function $\psi(\theta)$,

$$|\psi(\theta)|^2 = \prod_j \{1 - 4p_j(1 - p_j) \sin^2(\theta/2)\}, \quad |\psi(\theta)| \leq e^{-2V\theta^2/\pi^2} \quad (|\theta| \leq \pi).$$

Finite total mean ensures the sum is almost surely finite and allows passage from finite products to this limit. On a fixed small interval $|\theta| \leq \theta_0$, put $D_p(\theta) = 1 - p + pe^{i\theta}$ and $\ell_p(\theta) = \log D_p(\theta) - ip\theta$, with the branch from zero. The denominator obeys $|D_p| \geq \cos(\theta_0/2) > 0$ uniformly in p , and

$$\ell_p'''(\theta) = -\frac{ip(1 - p)e^{i\theta}(1 - p - pe^{i\theta})}{D_p(\theta)^3}.$$

Thus $|\ell_p''| \leq Bp(1 - p)$ with an absolute B , while $\ell_p(0) = \ell_p'(0) = 0$ and $\ell_p''(0) = -p(1 - p)$. The centered logarithms are summable, and Taylor's theorem gives

$$\log \psi(\theta) = -V\theta^2/2 + R(\theta), \quad |R(\theta)| \leq B_1 V |\theta|^3.$$

Choose θ_0 so that $B_1\theta_0 \leq 1/4$. The exponential difference identity gives

$$|\psi(\theta) - e^{-V\theta^2/2}| \leq B_1V|\theta|^3e^{-V\theta^2/4}.$$

Its integral is $O(V^{-1})$. The remaining arcs and the omitted Gaussian tails are exponentially small by the modulus bound. Fourier inversion at the integer mean proves (69). No logarithm near $\theta = \pi$ is used.

The exact tilting identity, followed by (56), (68) and (69), now gives

$$\begin{aligned} p(m) &= \frac{C(r_m)}{C(1)r_m^m} \mathbb{P}_{r_m}(J = m) \\ &= \frac{e^{\mathcal{A}k_m} k_m^{-2m-1}}{2\sqrt{2}\pi^{3/2}C(1)\sqrt{m}} \{1 + O(m^{-1/2})\}. \end{aligned}$$

Set $k_0 = 2m/\mathcal{A}$. Since $\Delta = k_m - k_0 = O(1)$,

$$\mathcal{A}\Delta - (2m+1)\log(1 + \Delta/k_0) = O(m^{-1}).$$

The order-one linear terms cancel exactly. Replacing k_m by k_0 therefore changes the answer only by $1 + O(m^{-1})$. Simplification proves (64). Stirling's formula in the form

$$(2m+1)! = 4\sqrt{\pi} m^{3/2} (2m/e)^{2m} \{1 + O(m^{-1})\}$$

proves the equivalent factorial statement (63); its denominator is $\sqrt{2}$ times π , not $\sqrt{2\pi}$.

Dividing the mass estimates at consecutive integers gives

$$\frac{p(m+1)}{p(m)} = \frac{\mathcal{A}^2}{4m^2} \{1 + O(m^{-1/2})\}.$$

Consequently all sufficiently late consecutive ratios are bounded by B_2/m^2 for a fixed B_2 . Summing the resulting geometric majorant shows $\sum_{j \geq 2} p(m+j)/p(m) = O(m^{-4})$. Adding the $j = 1$ term proves (65) (and even an $O(m^{-5/2})$ remainder there). This does not improve the relative error of the mass formula itself. $\square$

Corollary 10.4. *Let $q_m = \mathbb{P}(J \geq m)$ and define the small-tail threshold explicitly by*

$$n(y) = \min\{m \geq 0 : q_m \leq y\}, \quad y \downarrow 0.$$

Put $T = \log(1/y)$ and

$$H(x) = 2x \log \frac{2x}{e\mathcal{A}} + \frac{3}{2} \log x - \log K_{\text{tail}}.$$

Let x be the unique eventual solution of $H(x) = T$. Constants $D > 0$ and $y_* > 0$ exist such that for $0 < y < y_*$,

$$\left\lfloor x - \frac{Dx^{-1/2}}{\log x} \right\rfloor \leq n(y) \leq \left\lceil x + \frac{Dx^{-1/2}}{\log x} \right\rceil. \quad (70)$$

A smooth approximation is

$$\begin{aligned} x_0 &= \frac{T}{2W_0(T/(e\mathcal{A}))}, & x_1 &= x_0 - \frac{(3/2)\log x_0 - \log K_{\text{tail}}}{2\log(2x_0/\mathcal{A})}, \\ x &= x_1 + O\left(\frac{1}{x_0 \log x_0}\right). \end{aligned} \quad (71)$$

The correction $x_1 - x_0$ tends to $-3/4$.

Proof. The theorem gives $-\log q_m = H(m) + O(m^{-1/2})$ and $H'(x) = 2\log(2x/\mathcal{A}) + 3/(2x) \sim 2\log x$. Since every mass is positive, q_m decreases strictly. Choose D large enough that the carrier change over the real displacement in (70) dominates the logarithmic error. At the integer ceiling of the upper endpoint, $q_m \leq y$; at the integer immediately below the lower ceiling, $q_m > y$. These integers lie within $1 + o(1)$ of x , so the errors are uniformly $O(x^{-1/2})$. Monotonicity gives both bounds. The formulas in (71) invert the leading term and apply one Newton correction; the remaining carrier residual is $O(1/x_0)$, giving the stated inverse error. $\square$

The tilts used in the proof grow as $r_m \sim 4m^2/\mathcal{A}^2$, outside every fixed compact mark set. The argument uses the exact limiting product and Theorem 10.1, never a substitution into the finite- N expansion. It establishes neither simultaneous finite-size rare tails nor all fixed tail corrections. The inverse constants are non-effective and a single unconditional ceiling remains unjustified. Exponential tilting, the Bernoulli local estimate, convex secants and the monotone-density argument are classical methods, specialized here with the exact endpoint constant.

11 Simple limiting zeros and finite nonreal zeros

The determinant formula already puts every zero of C on the negative real axis. Simplicity follows from the special Green operator.

Theorem 11.1. *Every positive eigenvalue κ_j of K is simple. Hence the zeros $-1/\kappa_j$ of C are all simple and negative. For each fixed bounded disk, all zeros of P_{N+2} in that disk are real and simple for all sufficiently large N . In particular, all nonreal zeros of the finite polynomials escape every bounded disk as $N \rightarrow \infty$.*

Proof. Put $k_0(t) = \log(\rho/t)$. This belongs to $L^2(\mu)$: near zero its squared logarithm is integrable, and near ρ it vanishes quadratically against a weight with only a simple pole. Since $0 \leq k(s, t) \leq k_0(t)$, dominated convergence shows that $s \mapsto k(s, \cdot)$ extends continuously as an $L^2(\mu)$ -valued function to $[0, \rho]$, with endpoint values k_0 and zero.

If $Kf = \kappa f$ with $\kappa > 0$, the integral representation therefore supplies a continuous representative of f on $[0, \rho]$, with $f(\rho) = 0$. For $0 < s < \rho$, split the integral at s and differentiate; the matching boundary terms cancel, leaving

$$f'(s) = -\frac{1}{\kappa s} \int_0^s E(t)f(t) dt. \quad (72)$$

Continuity of f and E at zero makes the right-hand side bounded near zero. Integrating again gives

$$f(s) = f(0) - \kappa^{-1} \int_0^s E(t)f(t) \log \frac{s}{t} dt. \quad (73)$$

If $f(0) = 0$, iteration of the Volterra bound forces $f = 0$. Thus evaluation at zero is injective on each eigenspace, which must be one dimensional. Since K is self-adjoint, its eigenvalues have no additional algebraic multiplicity. The trace-class product now gives simple zeros of C .

Theorem 5.1 gives $R_N \rightarrow C$ locally uniformly. Choose a larger disk containing the fixed disk whose boundary contains no zero of C . It contains only finitely many limiting zeros. Place small disjoint disks, invariant under conjugation, around these simple real zeros. Rouché's theorem gives exactly one zero of R_N in each small disk for large N , counted with multiplicity. The real coefficients force that unique zero to be real, and its multiplicity is one. On the remaining compact portion C is bounded away from zero, so uniform convergence excludes any further zeros. The nonzero normalization in (24) does not change zeros of P_{N+2} . This proves the assertion, including when the original fixed disk has a limiting zero on its boundary. $\square$

The theorem is a bounded-region statement. It neither proves eventual complete real-rootedness nor bounds the rate of escape of nonreal roots. In fact complete real-rootedness fails at a modest finite index:

$$\begin{aligned}
 P_{10}(\lambda) = & \lambda^8 + 204\lambda^7 + 13230\lambda^6 + 366404\lambda^5 \\
 & + 4948521\lambda^4 + 33906984\lambda^3 + 112276976\lambda^2 \\
 & + 152976000\lambda + 55843200.
 \end{aligned} \tag{74}$$

An exact Sturm chain, starting with P_{10} and its derivative and taking negative Euclidean remainders, has degrees 8, 7, 6, 5, 4, 3, 2, 1, 0. Its signs at infinity are

	8	7	6	5	4	3	2	1	0	variations
$-\infty$	+	-	+	-	+	-	+	+	-	7
$+\infty$	+	+	+	+	+	+	+	-	-	1

so it has exactly six real roots. Its exact gcd with its derivative is one, and its positive coefficients exclude nonnegative real roots. Thus six roots are negative real and two form a nonreal conjugate pair. The accompanying exact code verifies the chain over the rationals; approximate roots are not used to establish the conclusion. This counterexample also rules out a finite Poisson-binomial interpretation for the stated marking.

12 Computational evidence and reproducibility

The proofs above are conventional mathematical proofs, not formal proof-assistant developments. The accompanying source archive separates exact finite checks, the amplitude certificate and numerical diagnostics.

The standard-library core regenerates the original integer triangle and its marked polynomial version, independently evaluates the formal ODE, and checks agreement with the inspected OEIS prefix. It constructs Frobenius and correction coefficients using exact rational polynomial arithmetic, verifies the displayed marked and scalar corrections, and performs the exact Sturm calculation for (74). It also checks the rational Taylor-tail bound used in (35). All correctness checks raise explicit exceptions and remain active under optimized Python.

The optional environment contains the separately pinned symbolic and multiprecision programs. These reproduce the 420-term Wronskian diagnostic, the interval enclosure, and high-order numerical comparisons. Optional dependencies are not needed to run the exact core or to rebuild the PDF from its supplied L^AT_EX source. The package README gives the precise commands and output inventory. The build disables shell escape, fixes archival timestamps and checks the source manifest; third-party papers are cited and are not redistributed.

For scale, the original scalar diagnostics report the following absolute errors of the normalized twelve-term falling-factorial approximation. They compare the exact integer diagonal with the high-precision carrier in (28):

N	Absolute normalized error
40	$3.70 \cdot 10^{-12}$
80	$1.72 \cdot 10^{-16}$
160	$1.35 \cdot 10^{-20}$
320	$1.33 \cdot 10^{-24}$
640	$1.46 \cdot 10^{-28}$
800	$7.86 \cdot 10^{-30}$

These are diagnostics, not uniform error certificates. At small N , adding many terms can worsen the approximation; the theorem is fixed-order asymptotics. Exact finite checks support the formulas but cannot replace the analytic continuation, parameter uniformity, trace-class or inverse arguments.

13 Source limits and further directions

The source comparison checked the official OEIS records, the classical transform paper, standard singularity-analysis sources, the relevant HKPV spectral theorem, and the cited classical Bessel formulas. All 21 displayed A386381 terms were compared with independent recurrence values. Access to the linked b-file did not succeed during that comparison; no assertion about its farther terms is based on that failed retrieval.

[write] **Added 7 October 2026.** The b-file (Kotěšovec, $0 \leq n \leq 264$) was fetched by the write on 7 October 2026: all 265 terms agree with the write's exact recomputation from (1) (Remark 1.1).

A bounded comparison of existing report titles and full relevant README files located no report on this particular modified Entringer endpoint problem. Related material on central Poupard numbers, zigzag Eulerian spectra, and surjection diagonals uses different objects or only shares classical methods. Search misses, including unreliable fuzzy searches among generically titled documents, are not evidence of worldwide novelty or exhaustive non-overlap. This report makes no claim to settle an explicitly posed open problem.

Several concrete extensions remain outside the established results:

1. **Effective remainders and threshold algorithms.** The displayed amplitude certificate is explicit, but the transfer and inverse constants are not. Quantitative contour bounds could turn the inverse bracket into a certified finite-input algorithm, with exact recurrence evaluation only near its integer boundaries.
2. **Spectral computation.** Certified eigenvalue enclosures for K would provide rigorous numerical values for μ_J , v_J and the limiting zeros. Any quadrature should preserve or control the endpoint singular weight rather than assume the underlying measure is finite.
3. **Nonreal root geometry.** The escape theorem gives no rate or limiting shape. The degree-eight counterexample shows why global root questions require separate analysis; it does not settle whether all sufficiently late finite polynomials become real-rooted.
4. **Large marks and rare counts.** The real-positive endpoint equivalent and limiting-tail theorem leave their higher corrections, uniform complex sectors, spectral counting, and simultaneous finite-size rare tails open here. Such finite-size results do not follow just by substituting a growing mark into the compact-mark coefficient expansion.
5. **Farther singularities.** Continuation toward -3ρ and beyond may reveal exponentially smaller contributions and further connection data. The present dominant expansion alone does not specify a complete exponential transseries.
6. **A combinatorial model for the mark.** The recursive coefficient statistic is exact. A natural independently recognizable family with that statistic would require a new construction or bijection, rather than an interpretation inferred only from coefficient positivity.

[write] **Added 7 October 2026, under Vladimir's standing rule of 4 October 2026.** The six directions above are the source's open questions, credited to it and kept as stated. The write found no further unproved claim in the source and no wrong one: every statement proved above has a complete argument, and

the remaining assertions are scoped as non-claims (Section 1.1). For item 4, the diagnostic after the proof of Theorem 10.1 suggests a first large-mark correction coefficient near 0.0278; it is not a theorem here.

The central conclusions are already exact: the posted scalar amplitude is a positive connection value, every fixed correction has an explicit recurrence, and the marked endpoint is a trace-class determinant whose simple spectrum explains the limiting Bernoulli law and the bounded-region zero behavior.

References

- [1] OEIS Foundation Inc., *A386363*, modified Entringer triangle, entry attributed to Mikhail Kurkov, 19 July 2025. Inspected 3 October 2026 (and by the write on 7 October 2026, revision #24). [OEIS record](#).
- [2] OEIS Foundation Inc., *A386381*, diagonal of A386363. Leading asymptotic and numerical amplitude attributed to Václav Kotěšovec, 2 September 2025. Inspected 3 October 2026 (and by the write on 7 October 2026, revision #27, with its b-file). [OEIS record](#).
- [3] J. Millar, N. J. A. Sloane and N. E. Young, *A new operation on sequences: the boustrophedon transform*, Journal of Combinatorial Theory, Series A **76** (1996), 44–54, Theorem 1. [Author-hosted paper](#).
- [4] P. Flajolet and A. Odlyzko, *Singularity analysis of generating functions*, SIAM Journal on Discrete Mathematics **3** (1990), 216–240. [Author-hosted paper](#).
- [5] P. Flajolet and R. Sedgewick, *Analytic Combinatorics*, Cambridge University Press, 2009, Chapter VI. [Author-hosted text](#).
- [6] J. B. Hough, M. Krishnapur, Y. Peres and B. Virág, *Determinantal processes and independence*, Probability Surveys **3** (2006), 206–229, Theorem 7 and its proof. [doi:10.1214/154957806000000078](https://doi.org/10.1214/154957806000000078).
- [7] NIST, *Digital Library of Mathematical Functions*, Chapter 10: [10.25.2](#) and [10.32.9](#) (definitions and positivity), [10.28.2](#) (Wronskian), [10.30](#) (endpoint limits), and [10.40.1–4](#) (large-argument function and derivative expansions).